

Entropy Collapse and Concentration Growth as Structural Signatures of Anomalous Behavior: Evidence from Financial Fraud and Blockchain Transaction Networks

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Abstract

Modern fraud and anomaly detection systems are commonly optimized for predictive performance, yet their internal decision mechanisms often provide limited insight into the structural organization of anomalous behavior. This paper develops and evaluates an interpretable descriptor framework derived from Arithmetic Power Geometry (APG) for studying entropy, concentration, and exponent-defect patterns in financial and blockchain transaction data. The central question is not whether APG outperforms state-of-the-art classifiers, but whether APG descriptors expose stable structural tendencies that help explain anomalous activity. We formulate two empirical principles: the Entropy Collapse Principle, which states that anomalous behavior may occupy fewer effective dimensions than normal behavior, and the Concentration Growth Principle, which states that anomalous behavior may be dominated by fewer structural components. Experiments are reported on the Credit Card Fraud dataset and the Elliptic Bitcoin transaction network. In the credit-card benchmark, fraud transactions show lower normalized APG entropy than normal transactions (0.6605 versus 0.7274) and higher concentration (0.3249 versus 0.2696), with statistically significant descriptor differences. In the Elliptic graph setting, graph-derived APG descriptors also show entropy collapse and concentration growth for illicit nodes; however, APG descriptors computed directly on raw Elliptic transaction features show the opposite directional pattern, demonstrating that the entropy-collapse hypothesis is representation-dependent rather than universal in all feature spaces. Predictive experiments show that APG descriptors provide complementary information to simple baselines but do not replace strong supervised classifiers such as logistic regression, XGBoost, or graph neural networks. The study positions APG as an interpretable structural and feature-engineering framework for anomaly analysis, rather than as a standalone state-of-the-art fraud detector.

Keywords: anomaly detection; fraud detection; blockchain analytics; explainable artificial intelligence; entropy; concentration; Arithmetic Power Geometry; graph anomaly detection.

1 Introduction

Fraud detection, anti-money-laundering analysis, and cybersecurity monitoring have increasingly become machine-learning problems. Classical linear models, tree ensembles, gradient-boosting systems, and graph neural networks now provide strong predictive baselines for tabular and graph-structured anomaly detection (Chen and Guestrin, 2016; Kho and Veal, 2020; Kipf and Welling, 2017; Hamilton et al., 2017; Velickovic et al., 2018; Weber et al., 2019). Yet the strongest models are often difficult to interpret. In high-stakes domains such as financial fraud, blockchain monitoring, and cyber defense, an analyst frequently needs more than a probability score: the analyst needs a structural explanation of why a transaction, node, or network neighborhood appears suspicious (Rudin, 2019; Lundberg and Lee, 2017; Molnar, 2022).

This paper investigates whether anomalous behavior exhibits measurable information-geometric structure. In particular, we study whether fraud and illicit activity are associated with entropy reduction and concentration increase under descriptor maps derived from Arithmetic Power Geometry (APG). APG is a local exponent-deformation framework in which normalized coordinate-weight distributions, entropy, concentration, and power-defect quantities describe how feature mass is organized across dimensions (Akhtar, 2026a,b,c,d). Previous APG work established that Shannon entropy governs local exponent sensitivity near the Euclidean reference exponent and that APG defect functionals can be bounded using Shannon–Renyi information structure. The present work asks whether these descriptors are useful beyond mathematical deformation theory, specifically in empirical anomaly analysis.

The motivation is simple. In many anomalous systems, behavior may become less diverse and more concentrated. A fraudster may use a narrower set of behavioral pathways than a legitimate user; an illicit blockchain transaction may occupy a concentrated graph neighborhood; an intrusion may collapse traffic diversity into a dominant protocol, port, or flow pattern. These observations motivate two empirical principles. First, anomalous behavior may show *entropy collapse*. Second, anomalous behavior may show *concentration growth*. APG provides a compact way to quantify both effects through a normalized squared-coordinate distribution.

The primary contribution of this work is not the development of a superior classifier. The reported results do not support such a claim. Instead, this study introduces an interpretable anomaly-analysis framework and systematically evaluates where APG descriptors reveal meaningful structural differences, where they provide complementary predictive information, and where their directional interpretation depends on the chosen representation. The APG-only score is not competitive with state-of-the-art fraud-detection models on the Credit Card Fraud benchmark, and graph neural networks remain the leading approach for blockchain transaction classification. Nevertheless, APG descriptors exhibit statistically significant separation patterns and provide useful structural information that can enhance simple graph-feature baselines in the Elliptic experiment. These findings position APG as an interpretable structural-descriptor and feature-engineering framework rather than as a replacement for established supervised learning methods.

The paper makes the following contributions:

- C1.** It formalizes APG entropy, normalized entropy, concentration, and defect descriptors for anomaly analysis.
- C2.** It states the Entropy Collapse Principle and Concentration Growth Principle as

empirical hypotheses rather than universal theorems.

- C3.** It evaluates APG descriptors on the Credit Card Fraud dataset and the Elliptic Bitcoin transaction network.
- C4.** It compares APG-only, baseline-only, and baseline-plus-APG models using ROC-AUC, PR-AUC, F1, precision, recall, and Matthews correlation coefficient.
- C5.** It reports a representation-dependent limitation: graph-derived Elliptic descriptors support entropy collapse, whereas raw Elliptic feature APG descriptors show higher entropy for illicit nodes.
- C6.** It positions APG as an explainable descriptor and feature-engineering layer rather than as a replacement for XGBoost, logistic regression, or graph neural networks.

2 Research Questions and Scope

This manuscript is organized around four research questions rather than a single classifier-optimization objective.

RQ1. Do APG descriptors separate anomalous and normal observations in real fraud and blockchain data?

RQ2. When APG descriptors separate classes, do they support the directional hypotheses of entropy collapse and concentration growth?

RQ3. Do APG descriptors provide complementary predictive information when added to conventional baseline features?

RQ4. Where does the APG interpretation fail, and what does that failure imply for future APG anomaly research?

These questions are deliberately conservative. The present paper does not claim that APG is a state-of-the-art fraud detector. The paper also does not claim that entropy collapse is a mathematical law that must hold for all data representations. Instead, it treats APG as an interpretable descriptor map whose empirical behavior must be tested representation by representation. This framing is important because anomaly detection is sensitive to preprocessing, feature construction, graph sampling, temporal splits, class imbalance, and threshold selection.

The intended contribution is therefore closer to a structural science of anomaly signatures than to a competition-style machine-learning system. In this respect, the paper follows an explanatory route: it asks what changes in the geometry of the data when a transaction becomes fraudulent or illicit. A classifier can identify a label, but a descriptor framework can help state what kind of structural change has occurred.

2.1 Supported and Unsupported Claims

For clarity, Table 1 distinguishes conclusions that are directly supported by the reported experiments from broader claims that are not established by the present evidence. The results demonstrate that APG descriptors reveal statistically significant entropy, concentration, and defect differences and may provide complementary features for baseline models. However, the present experiments do not support claims that APG replaces modern supervised learning methods, consistently improves all classifiers, or constitutes a

state-of-the-art fraud-detection framework. The purpose of this distinction is to ensure that the empirical findings are interpreted within the scope of the available evidence.

Table 1: Supported and unsupported claims in the present study.

Supported by the present results	Not supported by the present results
Credit-card fraud exhibits lower APG entropy and higher APG concentration than normal transactions.	APG is a state-of-the-art credit-card fraud detector.
Graph-derived Elliptic descriptors exhibit lower entropy and higher concentration for illicit nodes.	Entropy collapse holds for every possible representation of the Elliptic dataset.
APG descriptors can provide complementary information for selected baseline models.	APG consistently improves all strong classifiers.
APG descriptors are interpretable and statistically significant across the reported experiments.	APG replaces XGBoost, GCN, GraphSAGE, GAT, or other established predictive models.

3 Related Work

3.1 Classical and machine-learning fraud detection

Credit-card fraud detection is characterized by severe class imbalance, concept drift, and asymmetric error costs (Dal Pozzolo et al., 2015; Carcillo et al., 2021; Kho and Vea, 2020). The public European cardholder dataset made available by Dal Pozzolo et al. (2015) has become a common benchmark because it contains 284,807 transactions with only 492 fraud cases. Classical logistic regression remains useful as an interpretable baseline, while random forests and gradient-boosting systems often provide strong predictive performance (Breiman, 2001; Chen and Guestrin, 2016). Methods such as SMOTE have also been widely used for imbalanced classification (Chawla et al., 2002), although leakage and resampling design must be handled carefully.

Recent fraud-detection research increasingly combines ensemble learning, deep learning, representation learning, and explainable AI. However, many accuracy-focused papers provide limited structural interpretation beyond feature importance or post-hoc explanation. This paper differs by emphasizing a descriptor-level structural hypothesis: whether fraud exhibits entropy collapse and concentration growth in APG space.

3.2 Blockchain anti-money-laundering and graph learning

Blockchain transaction analysis is naturally graph-structured. The Elliptic Bitcoin dataset introduced a labeled transaction graph for anti-money-laundering research and helped bring graph convolutional networks into financial crime detection (Weber et al., 2019). Graph neural networks such as GCN, GraphSAGE, and GAT have become standard baselines for node classification and transaction-network analysis (Kipf and Welling, 2017; Hamilton et al., 2017; Velickovic et al., 2018). Recent work continues to extend blockchain AML modeling through temporal graph learning, attention mechanisms, and larger money-laundering graph datasets (Liu et al., 2024; Li et al., 2025; Lin et al., 2026).

APG does not replace graph neural networks. Instead, it can provide graph-derived structural descriptors, such as neighborhood entropy and concentration, that may be used either for interpretation or as additional node features.

3.3 Entropy and information-theoretic anomaly detection

Entropy has a long history as a measure of uncertainty and distributional spread (Shannon, 1948; Renyi, 1961; Cover and Thomas, 2006). In anomaly detection, entropy-based descriptors are used to quantify irregularity, uncertainty, diversity, and concentration in feature distributions, traffic flows, and graph neighborhoods. Information-theoretic anomaly methods are attractive because they can provide compact interpretable statistics rather than opaque latent embeddings.

The APG contribution is not merely to compute Shannon entropy. APG defines entropy on the squared-coordinate weight distribution and couples entropy to power-defect descriptors. This produces a joint entropy–concentration–defect representation that is compatible with local exponent-deformation theory.

3.4 Explainable artificial intelligence

Explainable AI methods such as LIME and SHAP provide local explanations for model predictions (Ribeiro et al., 2016; Lundberg and Lee, 2017). Although useful, post-hoc explanation methods often explain a trained model rather than the structure of the underlying phenomenon. APG descriptors instead describe the input geometry itself. This makes them complementary to model-explanation tools. For example, SHAP may identify which variables drive an XGBoost fraud score, while APG entropy and concentration summarize how the transaction’s feature mass is organized.

3.5 Arithmetic Power Geometry

Arithmetic Power Geometry was introduced as a local exponent-deformation framework studying how normalized closure defects evolve as the exponent moves away from a Euclidean reference target (Akhtar, 2026a). APG II developed an information-geometric extension with stabilized tensor formulations and coordinate-stability principles (Akhtar, 2026b). The Shannon–Renyi control manuscript established bridge identities and entropy bounds for APG closure defects (Akhtar, 2026c). APG-AMS then applied APG entropy as an entropy-guided screening mechanism for adaptive metric search (Akhtar, 2026d). The present paper is an applied extension: it tests whether APG descriptors expose interpretable anomaly structure in financial and blockchain data.

4 APG Descriptor Framework

Let $x = (x_1, \dots, x_d) \in \mathbb{R}_{>0}^d$ denote a positive feature vector. In practice, real-valued transaction features are first robustly scaled using training data and then mapped to positive coordinates. APG assigns each coordinate a squared-mass weight.

Definition 1 (APG weight distribution). *For $x \in \mathbb{R}_{>0}^d$, define*

$$W_i(x) = \frac{x_i^2}{\sum_{j=1}^d x_j^2}, \quad i = 1, \dots, d. \quad (1)$$

Then $W(x) = (W_1(x), \dots, W_d(x))$ is a probability distribution.

Definition 2 (APG entropy). *The APG entropy of x is*

$$H(W) = - \sum_{i=1}^d W_i \log W_i. \quad (2)$$

Definition 3 (Normalized APG entropy). *For $d > 1$, the normalized APG entropy is*

$$H_N(W) = \frac{H(W)}{\log d}. \quad (3)$$

This maps the entropy to $[0, 1]$ when all weights are positive.

Definition 4 (APG concentration). *The APG concentration of x is*

$$C(W) = \max_{1 \leq i \leq d} W_i. \quad (4)$$

Definition 5 (APG power-defect descriptor). *For $p > 0$, define*

$$D_{\text{APG}}(p; x) = \left| 1 - \frac{\sum_{i=1}^d x_i^p}{\left(\sum_{i=1}^d x_i^2\right)^{p/2}} \right|. \quad (5)$$

The defect in Equation (5) measures how the normalized power sum changes when the exponent departs from the Euclidean reference exponent $p = 2$. The APG Shannon–Renyi control theorem motivates the use of entropy as a deformation-sensitivity descriptor. If

$$D(t) = 1 - \sum_i W_i^{t/2}, \quad t \geq 2, \quad (6)$$

then the APG Shannon–Renyi bound gives

$$D(t) \leq 1 - \exp\left(-\frac{t-2}{2}H(W)\right) \leq \frac{t-2}{2}H(W). \quad (7)$$

Equation (7) is used here as theoretical motivation for entropy-based descriptors; it is not used to claim that APG entropy alone is an optimal classifier.

5 Entropy Collapse and Concentration Growth

Hypothesis 1 (Entropy Collapse Principle). *In a descriptor representation that captures anomaly-relevant structure, anomalous observations tend to have lower APG entropy than normal observations:*

$$H_{\text{anomaly}} < H_{\text{normal}}. \quad (8)$$

Hypothesis 2 (Concentration Growth Principle). *In a descriptor representation that captures anomaly-relevant structure, anomalous observations tend to have higher APG concentration than normal observations:*

$$C_{\text{anomaly}} > C_{\text{normal}}. \quad (9)$$

Remark 1. *These are empirical principles, not universal mathematical theorems. Their validity can depend on the feature representation. The Elliptic experiments in this paper explicitly demonstrate this dependence: graph-derived APG descriptors support both principles, whereas raw Elliptic feature APG descriptors do not.*

6 Mathematical Interpretation of the APG Descriptors

The APG descriptor family has a simple interpretation. The weight distribution in Equation (1) converts a positive feature vector into a probability distribution over coordinates. Entropy then measures the effective spread of the squared feature mass, while concentration measures the dominance of the largest coordinate. Defect terms measure how the normalized power sum changes when the exponent is moved away from the Euclidean reference value.

Proposition 1 (Entropy maximum). *For any APG weight distribution $W = (W_1, \dots, W_d)$,*

$$0 \leq H(W) \leq \log d. \quad (10)$$

The upper bound is attained exactly when $W_i = 1/d$ for all i , and the lower limit is approached when a single coordinate carries all mass.

This elementary fact explains why normalized entropy is useful for comparing vectors of the same dimension. High entropy indicates that squared feature mass is distributed broadly across coordinates. Low entropy indicates that feature mass has collapsed into fewer coordinates.

Proposition 2 (Concentration lower bound). *For any APG weight distribution,*

$$\frac{1}{d} \leq C(W) \leq 1. \quad (11)$$

The lower bound is attained at the uniform distribution, and the upper limit is approached by a fully concentrated distribution.

The two propositions show why entropy collapse and concentration growth are related but not identical. Entropy summarizes the full distribution of feature mass. Concentration records only the largest coordinate weight. A transaction can have the same maximum weight as another transaction while having different tail distribution over the remaining coordinates. Therefore both descriptors are retained.

6.1 Why APG defects are included

Entropy and concentration describe the coordinate-weight distribution. APG defects add an exponent-deformation view.

For $p < 2$, the APG defect

$$D_{\text{APG}}(p; x) = \left| 1 - \frac{\sum_{i=1}^d x_i^p}{\left(\sum_{i=1}^d x_i^2\right)^{p/2}} \right| = \left| 1 - \sum_{i=1}^d W_i^{p/2} \right|$$

measures sub-Euclidean exponent deformation through the absolute deviation of the normalized power sum from its Euclidean reference value. Because $p/2 < 1$, the quantity $1 - \sum_i W_i^{p/2}$ is generally non-positive, and the absolute value is required to obtain a nonnegative descriptor.

For $p \geq 2$, the normalized power-sum defect

$$D(p) = 1 - \sum_{i=1}^d W_i^{p/2}$$

is nonnegative and admits Shannon–Rényi control. The set $p \in \{1.5, 3, 4\}$ is interpreted as a small, interpretable exponent panel that samples both sub-Euclidean and super-Euclidean deformation behavior. These exponents were selected to provide representative coverage of different deformation regimes rather than through an optimized hyperparameter search.

The Shannon–Rényi control relation in Equation (7) provides the theoretical connection between entropy and exponent deformation. It says that, near and above the Euclidean reference exponent, entropy bounds the normalized power-sum defect. In an applied anomaly setting, this motivates the use of entropy as a deformation-sensitivity descriptor. This does not imply that a low-entropy observation must be fraudulent. The Shannon–Rényi bound is invoked only for exponents $p \geq 2$. Accordingly, $D_{\text{APG}}(1.5)$ is treated as an empirical sub-Euclidean deformation descriptor rather than as a quantity directly governed by the entropy bound.

6.2 Representation Dependence

Let ϕ be a feature map from an object space $\mathcal{X}$ into a positive coordinate space. APG descriptors are computed after the application of ϕ . Therefore APG is not invariant to arbitrary feature maps. If ϕ captures anomaly-relevant structure, entropy collapse may appear. If ϕ captures another aspect of the system, the direction may reverse. This is exactly what the Elliptic results show: graph-derived APG descriptors support entropy collapse, while raw transaction-feature APG descriptors do not.

This point is important for the interpretation of the present results. It protects against overclaiming and also opens a future research direction: discovering which representations reveal entropy collapse for which anomaly families.

7 Datasets

7.1 Credit Card Fraud dataset

The first dataset is the public Credit Card Fraud benchmark associated with European cardholders in September 2013 (Dal Pozzolo et al., 2015). The dataset contains anonymized PCA-derived transaction features, a transaction amount, time, and a binary fraud label. The experimental summary used in this paper is shown in Table 2.

7.2 Elliptic Bitcoin transaction network

The second dataset is the Elliptic Bitcoin transaction network, a graph benchmark for illicit transaction classification (Weber et al., 2019). Nodes represent Bitcoin transactions, edges represent transaction flow, and node labels indicate licit, illicit, or unknown status. The experiment in this paper uses only labeled nodes for supervised evaluation. The summary is shown in Table 3.

Table 2: Credit Card Fraud dataset summary.

Property	Value
Transactions	284,807
Features	30
Fraud cases	492
Normal cases	284,315
Fraud rate	0.1727%
Train/test split	70/30 stratified
Random state	42

Table 3: Elliptic Bitcoin dataset summary used in this experiment.

Property	Value
Total nodes	203,769
Edges	234,355
Labeled nodes	46,564
Illicit labeled nodes	4,545
Licit labeled nodes	42,019
Unknown nodes	157,205

8 Experimental Methodology

8.1 Preprocessing

For the tabular credit-card experiment, feature transformations were fitted using the training data only. Robust scaling was applied to reduce the influence of heavy-tailed values. Because APG weights require positive coordinates, the scaled feature vector z was mapped to positive coordinates by

$$x_i = |z_i| + \varepsilon, \quad (12)$$

where $\varepsilon > 0$ is a small numerical constant. APG descriptors were then computed from x .

For the Elliptic experiment, two descriptor families were evaluated. First, APG descriptors were computed on raw transaction-feature vectors after positivity transformation. Second, graph-derived APG descriptors were computed from local graph quantities, including degree and PageRank-derived structural weights. This distinction is essential because the directional entropy pattern differs across these representations.

8.2 APG feature construction

The following APG features were constructed:

$$\mathcal{F}_{\text{APG}} = \{H(W), H_N(W), C(W), D_{\text{APG}}(1.5), D_{\text{APG}}(3), D_{\text{APG}}(4)\}. \quad (13)$$

For the credit-card benchmark, an additional entropy-deviation feature and an aggregate APG score were also evaluated. The raw score was defined as a sum of defect terms, entropy deviation, and concentration. Because the raw score was inversely oriented on the training data, an oriented APG score was also reported.

8.3 Baselines and evaluation metrics

The credit-card experiment compares APG-only scoring with logistic regression, logistic regression plus APG features, XGBoost, and XGBoost plus APG features. XGBoost is included because gradient-boosted decision trees are strong tabular baselines (Chen and Guestrin, 2016). Logistic regression is included because it remains a transparent linear reference model.

The Elliptic experiment compares degree plus PageRank graph features, APG descriptors only, degree plus PageRank plus APG, graph-APG only, and degree plus PageRank plus graph-APG. Full GCN, GraphSAGE, and GAT models are discussed as established literature baselines rather than newly retrained models in this manuscript.

Evaluation metrics include ROC-AUC, PR-AUC, F1-score, precision, recall, specificity, accuracy, balanced accuracy, and Matthews correlation coefficient (MCC). Because fraud and illicit-transaction datasets are highly imbalanced, PR-AUC and MCC are emphasized alongside ROC-AUC.

9 Evaluation Metrics

Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives. The following metrics are used.

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \quad (14)$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (15)$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \quad (16)$$

ROC-AUC measures ranking quality across thresholds using true-positive and false-positive rates. PR-AUC is often more informative under extreme class imbalance because it focuses on the precision–recall tradeoff for the minority class. MCC is also reported because it uses all four confusion-matrix cells and is robust for imbalanced binary classification.

10 Algorithmic Summary

The overall APG descriptor construction and evaluation procedure is summarized in Algorithm 1.

Algorithm 1. APG Descriptor Construction and Evaluation

Input: Dataset $X = \{x_i\}_{i=1}^N$ with class labels y .

Output: APG descriptors and predictive evaluation results.

1. Partition the dataset into training and test sets using a stratified protocol to avoid label leakage.
2. Fit scaling and preprocessing parameters using the training data only.

3. Transform each feature vector into positive coordinates according to Equation (12).
4. Compute APG weights using Equation (1).
5. Compute entropy, normalized entropy, concentration, and defect descriptors using Equations (2)–(5).
6. Perform statistical comparisons of APG descriptors between normal and anomalous classes.
7. Train baseline predictive models with and without APG descriptors as additional features.
8. Evaluate predictive performance using ROC-AUC, PR-AUC, F1-score, precision, recall, and MCC.

The procedure separates descriptor construction from predictive modeling. Consequently, APG descriptors may be analyzed directly as structural indicators, incorporated as additional predictive features, or used as interpretable explanatory variables.

11 Credit Card Fraud Results

11.1 Predictive performance

Table 4 reports predictive performance on the stratified test split. The APG-only oriented score achieves ROC-AUC of 0.7372 but extremely low PR-AUC, indicating that it is not competitive as a standalone detector in this dataset. Logistic regression and XGBoost achieve much stronger predictive performance. Adding APG features to logistic regression slightly improves F1 and MCC, while adding APG to XGBoost does not improve the reported XGBoost test metrics.

Table 4: Credit Card Fraud benchmark results.

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	MCC
XGBoost	0.9709	0.7220	0.7786	0.8258	0.7365	0.7795
Logistic Regression	0.9680	0.7040	0.8100	0.8626	0.7635	0.8112
Logistic Regression + APG	0.9662	0.7013	0.8129	0.8692	0.7635	0.8144
APG-Fraud oriented score	0.7372	0.0034	0.0080	0.0040	0.6351	0.0339
APG-Fraud raw score	0.2628	0.0011	0.0034	0.0017	0.9527	-0.0014

The main conclusion from Table 4 is that APG descriptors should not be framed as a replacement for supervised fraud models. Their value is interpretability and complementary structure.

11.2 APG descriptor statistics

Table 5 reports class-wise APG feature means. Fraud transactions show lower APG entropy and normalized entropy than normal transactions. They also show higher maximum coordinate weight. This directly supports entropy collapse and concentration growth for the credit-card feature representation.

Table 5: Credit-card APG descriptor means by class.

APG feature	Normal mean	Fraud mean	Direction
Entropy $H(W)$	2.4742	2.2464	Fraud lower
Normalized entropy $H_N(W)$	0.7274	0.6605	Fraud lower
Max weight $C(W)$	0.2696	0.3249	Fraud higher
$D_{\text{APG}}(1.5)$	0.9393	0.8445	Fraud lower
$D_{\text{APG}}(3)$	0.6627	0.6192	Fraud lower
$D_{\text{APG}}(4)$	0.8537	0.8171	Fraud lower
APG score	2.8120	2.6924	Fraud lower

11.3 Statistical validation

Mann–Whitney and Kolmogorov–Smirnov tests were used to compare descriptor distributions. Table 6 reports p-values for major APG descriptors. All primary APG descriptors show statistically significant differences between fraud and normal transactions.

Table 6: Credit-card APG descriptor statistical tests.

APG feature	Mann–Whitney p-value	KS p-value
Entropy $H(W)$	1.96×10^{-20}	1.02×10^{-22}
Normalized entropy $H_N(W)$	1.96×10^{-20}	1.02×10^{-22}
Max weight $C(W)$	2.94×10^{-11}	7.96×10^{-12}
$D_{\text{APG}}(1.5)$	5.82×10^{-23}	2.14×10^{-26}
$D_{\text{APG}}(3)$	3.22×10^{-17}	8.65×10^{-19}
$D_{\text{APG}}(4)$	2.20×10^{-15}	3.60×10^{-17}
APG score	1.76×10^{-23}	2.98×10^{-24}

11.4 Credit-card interpretation

The credit-card experiment provides the cleanest support for the proposed APG principles. The normalized entropy reduction from 0.7274 to 0.6605 indicates that fraud transactions exhibit lower APG entropy and a more concentrated APG weight distribution. The concentration increase from 0.2696 to 0.3249 indicates that fraud transactions allocate more squared feature mass to a dominant coordinate. Together, these two patterns describe a shift from diffuse behavior to concentrated behavior.

The APG defects move in the lower direction for fraud in this dataset. This does not contradict the concentration interpretation. The defect values depend on the complete normalized power-sum geometry and the selected exponent. The important point is not that every APG descriptor must increase for fraud; rather, APG produces a multi-dimensional structural fingerprint whose components are statistically different across classes.

From a classifier perspective, the APG-only score has weak precision because the dataset is extremely imbalanced. A score can rank fraud cases above many normal cases and still produce low precision if the positive class is only 0.1727 percent of the data. This explains why ROC-AUC can be moderate while PR-AUC remains extremely small. For fraud detection, PR-AUC and MCC are therefore essential.

12 Elliptic Bitcoin Results

12.1 Predictive performance

Table 7 reports the Elliptic experiment. APG descriptors computed on raw transaction features achieve stronger ROC-AUC than degree plus PageRank alone. Combining degree plus PageRank with APG descriptors further improves ROC-AUC, PR-AUC, F1, and MCC over the weak graph-feature baseline. Graph-APG alone is weaker than raw-feature APG but provides interpretable structural descriptors.

Table 7: Elliptic Bitcoin supervised results on labeled nodes.

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	MCC
Degree + PageRank	0.5943	0.1155	0.1715	0.1146	0.3402	0.0366
APG descriptors only	0.7537	0.2023	0.3000	0.1885	0.7346	0.2399
Degree + PageRank + APG	0.7759	0.2424	0.3133	0.2002	0.7207	0.2543
Graph-APG only	0.6257	0.1344	0.2198	0.1435	0.4692	0.1059
Degree + PageRank + Graph-APG	0.6540	0.1428	0.2342	0.1435	0.6364	0.1350

These results support APG as a useful descriptor and feature-engineering layer for weak graph baselines. They do not show that APG outperforms graph neural networks. Published graph-learning methods on Elliptic and related blockchain AML tasks remain stronger references for predictive classification (Weber et al., 2019; Kipf and Welling, 2017; Hamilton et al., 2017; Velickovic et al., 2018; Li et al., 2025; Lin et al., 2026).

12.2 Raw-feature APG statistics

Table 8 reports APG statistics computed on raw Elliptic transaction features. In this representation, illicit nodes show higher entropy and lower concentration than licit nodes. This contradicts the entropy-collapse and concentration-growth directions and demonstrates that the hypothesis is not representation-invariant.

Table 8: Elliptic raw transaction-feature APG statistics.

Feature	Licit mean	Illicit mean	Difference (Illicit – Licit)	Mann–Whitney p-value
Entropy	0.8884	1.3840	0.4956	$< 10^{-300}$
Normalized entropy	0.1740	0.2711	0.0971	$< 10^{-300}$
Max weight	0.6853	0.5305	-0.1548	$< 10^{-300}$
$D_{\text{APG}}(1.5)$	0.3225	0.5151	0.1926	$< 10^{-300}$
$D_{\text{APG}}(3)$	0.2783	0.4170	0.1387	$< 10^{-300}$
$D_{\text{APG}}(4)$	0.4116	0.5966	0.1851	$< 10^{-300}$

12.3 Graph-feature APG statistics

Table 9 reports graph-derived APG statistics. In contrast with raw features, graph-derived descriptors show lower entropy and higher concentration for illicit nodes. This supports the claim that entropy collapse and concentration growth appear in the graph-structural representation.

Table 9: Elliptic graph-derived APG statistics.

Feature	Licit mean	Illicit mean	Difference	Mann–Whitney p-value
Entropy	0.6101	0.4890	-0.1210	3.08×10^{-120}
Max weight	0.7349	0.7998	0.0649	2.81×10^{-124}

The combined evidence from the Credit Card Fraud and Elliptic experiments reveals a representation-dependent pattern. Credit-card transactions and graph-derived Elliptic features exhibit lower entropy and higher concentration for anomalous observations, whereas raw Elliptic transaction features exhibit the opposite directional behavior. These findings suggest that entropy collapse and concentration growth should not be interpreted as universal properties of anomaly labels alone, but rather as representation-dependent structural signatures that emerge under informative feature representations.

Representation Dependence. The Elliptic experiments provide a direct counterexample to any claim that entropy collapse is representation invariant. Raw transaction features exhibit higher entropy for illicit nodes, whereas graph-derived features exhibit lower entropy. This suggests that entropy collapse should be interpreted as a representation-dependent structural phenomenon rather than a universal property of anomaly labels alone.

13 Comparison with Existing Methods

Table 10 compares the conceptual role of APG with common baseline families. Logistic regression is interpretable but not graph-aware. XGBoost is strong for tabular prediction but only partially interpretable without additional tools. GCN, GraphSAGE, and GAT exploit graph structure but are generally less transparent. APG is not the strongest classifier, but it explicitly exposes entropy, concentration, and defect structure.

Table 10: Conceptual comparison with existing method families.

Method	Predictive strength	Interpretable	Graph-aware	Entropy-based
Logistic regression	Medium	High	No	No
Random forest	High	Partial	No	No
XGBoost	High	Partial	No	No
GCN	High on graphs	Low	Yes	Implicit
GraphSAGE	High on graphs	Low	Yes	Implicit
GAT	High on graphs	Low	Yes	Implicit
SHAP/LIME	Explanation layer	High	Model-dependent	No
APG descriptors	Descriptor layer	High	Yes, if graph-derived	Yes

Table 11 summarizes claim-level differences between an accuracy-centered fraud paper and the present APG paper.

Table 11: Claim-level comparison.

Question	Accuracy-centered fraud paper	Present APG paper
Primary target	Maximize classification performance	Explain anomaly structure
Main object	Classifier	Descriptor family
Key evidence	ROC-AUC, PR-AUC, F1	Entropy, concentration, defect shifts plus metrics
Strong claim	Model improves prediction	APG reveals interpretable structural patterns
Main risk	Black-box behavior	Representation dependence
Recommended use	Deployed predictor	Explanation and feature-engineering layer

14 Discussion

The credit-card results strongly support the entropy-collapse and concentration-growth principles in the tested tabular representation. Fraud transactions show lower APG entropy and higher maximum coordinate weight than normal transactions. The p-values indicate that the differences are not random artifacts of the test split. However, the APG-only score performs poorly as a classifier because descriptor separation does not automatically imply high precision under extreme class imbalance.

The Elliptic results are more nuanced. Raw transaction-feature APG descriptors separate illicit and licit nodes and improve simple baseline performance, but they do not support entropy collapse directionally. Illicit nodes have higher raw-feature entropy and lower raw-feature concentration. In contrast, graph-derived APG descriptors show the expected entropy-collapse and concentration-growth pattern. This suggests that APG’s structural interpretation depends on whether the feature representation captures the relevant mechanism of anomaly. For blockchain illicit activity, graph structure may be more meaningful than raw transaction features.

This representation dependence is a central empirical finding of the present study. The results indicate that entropy collapse is not universal across arbitrary feature representations. Rather, it emerges when APG is applied to a representation that captures anomaly-relevant structural information. In the Credit Card Fraud dataset, the anonymized tabular feature representation supports entropy collapse and concentration growth. In the Elliptic dataset, the graph-derived representation exhibits the same pattern, whereas the raw transaction-feature representation exhibits a different directional behavior. These findings suggest that entropy collapse and concentration growth should be interpreted as representation-dependent structural signatures rather than universal properties of anomalous behavior.

15 Practical Implications

APG descriptors can be used in three practical ways. First, they can serve as interpretable audit variables. An analyst can inspect whether a transaction or node has unusually low entropy or unusually high concentration. Second, they can augment machine-learning

models as engineered features. The credit-card logistic regression experiment and the Elliptic degree-plus-PageRank experiment both show modest improvements when APG features are added. Third, they can guide future graph-learning models, where APG entropy and concentration may be appended to GNN node features or used as explanation signals.

16 Limitations

This paper has several limitations. First, APG does not outperform state-of-the-art classifiers in the reported experiments. Second, the Elliptic analysis is based on labeled nodes and simple graph-feature baselines; full GCN, GraphSAGE, and GAT retraining is left for future work. Third, the entropy-collapse principle is representation-dependent. Fourth, only two datasets are fully evaluated in this manuscript. Broader claims about universality would require additional datasets such as IEEE-CIS Fraud Detection, PaySim, UNSW-NB15, and CICIDS2017 (Lopez-Rojas et al., 2016; Moustafa and Slay, 2015; Sharafaldin et al., 2018). Fifth, the APG score formula is exploratory and should not be treated as a finalized fraud detector.

17 Extended Comparison with Existing Literature

Compared with XGBoost, APG is weaker as a predictive engine but stronger as a transparent descriptor family. XGBoost optimizes ensembles of boosted trees and is highly effective for tabular data (Chen and Guestrin, 2016). However, its explanations usually require feature-importance plots or post-hoc methods. APG descriptors are directly interpretable before any classifier is trained.

Compared with GCN, GraphSAGE, and GAT, APG is not a deep graph representation learner. GCN propagates features over graph neighborhoods using spectral or message-passing approximations (Kipf and Welling, 2017). GraphSAGE learns inductive aggregation functions (Hamilton et al., 2017). GAT uses attention weights over neighborhoods (Velićković et al., 2018). APG does not learn a neighborhood embedding. Instead, it summarizes a chosen representation through entropy, concentration, and defect. This makes APG simpler but less expressive.

Compared with SHAP and LIME, APG explains the object representation rather than the trained model. SHAP can explain why XGBoost predicts a transaction as fraud. APG can explain whether the transaction’s feature mass has become concentrated. These are complementary forms of explanation.

The best future positioning is therefore hybrid: APG descriptors should be added to strong learners and used as transparent anomaly-structure variables. A future APG-GNN model could append graph-APG entropy and concentration to learned node embeddings, allowing predictive power from graph neural networks and interpretability from APG descriptors.

18 Threats to Validity

18.1 Internal validity

The credit-card experiment uses a stratified 70/30 split. Although training-only scaling was used, additional validation should include repeated stratified splits and temporal validation. The Elliptic experiment uses labeled nodes only and simple graph-feature baselines. Full temporal graph splits should be tested in a stronger blockchain study.

18.2 External validity

Only two datasets are fully evaluated. A universal anomaly claim would require broader domains: simulated mobile-money fraud, large e-commerce fraud, intrusion detection, insider-threat logs, and temporal blockchain AML data. The present paper should be read as evidence and framework development, not final universality proof.

18.3 Construct validity

Entropy collapse depends on the representation. The raw Elliptic APG results show that an anomaly can have higher entropy in one feature space and lower entropy in another. Therefore APG descriptors should always be interpreted relative to the feature map.

18.4 Conclusion validity

Very small p-values indicate distributional differences, but statistical significance is not the same as practical detection performance. This is why the paper reports both p-values and predictive metrics.

19 Future Work

Future work should proceed in four directions. First, APG descriptors should be evaluated across additional tabular, graph, and cybersecurity datasets. Second, APG-GNN models should be developed by appending entropy, concentration, and defect descriptors to GCN, GraphSAGE, and GAT node representations. Third, temporal APG descriptors should be introduced to test whether entropy collapse occurs before an anomaly becomes visible to a classifier. Fourth, APG explanations should be combined with SHAP or LIME to produce both structural and model-based explanations.

20 Conclusion

This study introduced APG-Fraud, an interpretable anomaly-analysis framework based on entropy, concentration, and defect descriptors derived from Arithmetic Power Geometry. Across two distinct domains—financial fraud and blockchain transaction analysis—the proposed descriptors revealed consistent structural differences between anomalous and non-anomalous observations.

On the Credit Card Fraud dataset, fraudulent transactions exhibited lower entropy and higher concentration than normal transactions, supporting the Entropy Collapse Principle and Concentration Growth Principle. On the Elliptic Bitcoin transaction

network, graph-derived APG descriptors exhibited the same directional behavior, whereas APG descriptors computed directly from raw transaction features exhibited the opposite trend. This contrast demonstrates that the observed structural signatures are not invariant properties of anomaly labels alone, but depend critically on the representation through which the underlying system is analyzed.

Taken together, the results suggest that APG provides a useful structural-descriptor framework for interpretable anomaly analysis. Rather than competing with state-of-the-art predictive models such as XGBoost or graph neural networks, APG is best viewed as an explainable feature-engineering and structural-analysis layer that complements existing machine-learning systems. Its primary contribution lies in revealing how information concentration, entropy organization, and defect structure differ between normal and anomalous behavior.

Scientific Scope

The findings of this study support entropy collapse and concentration growth as empirically observable structural descriptors of anomalous behavior in selected feature representations. They should not be interpreted as universal laws of fraud, anomaly formation, or illicit activity. Instead, the evidence indicates that these phenomena are representation-dependent and emerge when the chosen feature space captures meaningful structural organization. Future research should investigate which feature representations, graph constructions, and domain-specific transformations most reliably reveal these signatures across financial, blockchain, cybersecurity, and other anomaly-detection settings. Such studies will help determine the broader applicability and theoretical limits of APG-based structural analysis.

Appendices

Additional supporting material is provided in the appendices. Tables 12 and 13 report complete confusion matrices, Table 14 presents extended performance metrics, Table 15 provides a reproducibility checklist, and Table 16 contains a citation audit. These materials are included to facilitate verification, replication, and future extension of the reported results.

A Complete Confusion-Matrix Details

Table 12: Credit-card confusion matrices at selected thresholds.

Model	TN	FP	FN	TP
XGBoost	85,272	23	39	109
Logistic Regression	85,277	18	35	113
Logistic Regression + APG	85,278	17	35	113
APG-Fraud oriented score	62,060	23,235	54	94
APG-Fraud raw score	3,471	81,824	7	141

Table 13: Elliptic confusion matrices.

Model	TN	FP	FN	TP
Degree + PageRank	9,022	3,584	900	464
APG descriptors only	8,292	4,314	362	1,002
Degree + PageRank + APG	8,678	3,928	381	983
Graph-APG only	8,787	3,819	724	640
Degree + PageRank + Graph-APG	7,424	5,182	496	868

B Extended Numerical Summary

Table 14: Credit-card full metric panel.

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Specificity	Accuracy	Balanced Acc.
XGBoost	0.9709	0.7220	0.7786	0.8258	0.7365	0.9997	0.9993	0.8681
Logistic Regression	0.9680	0.7040	0.8100	0.8626	0.7635	0.9998	0.9994	0.8817
Logistic Regression + APG	0.9662	0.7013	0.8129	0.8692	0.7635	0.9998	0.9994	0.8817
APG-Fraud oriented score	0.7372	0.0034	0.0080	0.0040	0.6351	0.7276	0.7274	0.6814
APG-Fraud raw score	0.2628	0.0011	0.0034	0.0017	0.9527	0.0407	0.0423	0.4967

C Future Validation Considerations

Future validation should include additional external datasets to assess the generalizability of the observed entropy-collapse and concentration-growth patterns across multiple anomaly-detection domains. IEEE-CIS Fraud Detection would test e-commerce fraud with richer categorical and identity variables. PaySim would test simulated mobile-money fraud. UNSW-NB15 and CICIDS2017 would test whether entropy collapse transfers from fraud to cybersecurity. Each dataset should be evaluated using repeated splits or temporal splits, and the paper should report whether entropy collapse appears in raw features, engineered features, graph features, or temporal features.

Future evaluations may compare:

1. Baseline model without APG.
2. Baseline model with APG descriptors.
3. APG-only interpretable score.
4. SHAP explanation of the baseline model.
5. APG structural explanation of the same observations.

This protocol would help distinguish predictive performance gains from improvements in interpretability and structural explanation.

D Reproducibility Summary

Table 15: Reproducibility checklist.

Item	Description
Credit-card split	70/30 stratified split, random state 42.
Credit-card prepro- cessing	Training-only robust scaling; positivity transform $x_i = z_i + \varepsilon$.
APG features	Entropy, normalized entropy, max weight, $D_{\text{APG}}(1.5)$, $D_{\text{APG}}(3)$, $D_{\text{APG}}(4)$.
Credit-card models	Logistic regression, logistic regression plus APG, XGBoost, APG raw and oriented scores.
Elliptic data	203,769 nodes; 234,355 edges; 46,564 labeled nodes.
Elliptic models	Degree plus PageRank, APG descriptors only, graph-APG descriptors, and combinations.
Metrics	ROC-AUC, PR-AUC, F1, precision, recall, MCC.

E Citation Audit

Table 16 summarizes the major reference groups and their role in the present study.

Table 16: Citation audit.

Group	In-text citation keys	Role
Credit-card fraud	dal2015calibrating, carcillo2021scarff, kho2020survey	Dataset and fraud-detection context.
Classical ML	breiman2001random, chen2016xgboost, chawla2002smote, pedregosa2011scikit	Baseline models and implementation context.
Graph learning	kipf2017gcn, hamilton2017graphsage, velickovic2018gat	GNN comparison.
Blockchain AML	weber2019aml, liu2024shape, li2025global, lin2026wavelet	Elliptic and recent AML comparison.
Information theory	shannon1948mathematical, renyi1961measures, cover2006elements	Entropy basis.
Explainability	ribeiro2016lime, lundberg2017shap, rudin2019stop, molnar2022interpretable	XAI comparison.

APG	akhtar2026apg1, akhtar2026apg2, akhtar2026renyi, akhtar2026ams	APG theoretical basis.
Future datasets	lopez2016paysim, moustafa2015unsw, sharafaldin2018cicids	Future validation.

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